25. The nonzero elements of $\mathbb{Z}_p$ form a group of order $p-1$ under multiplication modulo p, and the order of an element of a finite group divides the order of the group.

26. The elements of $\mathbb{Z}_n$ that are integers relative prime to n form a group of order $\phi(n)$ under multiplication modulo n, and the order of an element of a finite group divides the order of the group.

27. If $a^2 = 1$, then $a^2 - 1 = (a-1)(a+1) = 0$. Because a field has no divisors of 0, either $a - 1 = 0$ or $a + 1 = 0$. Thus either $a = 1$ or $a = p - 1$.

28. Because $\mathbb{Z}_p$ is a field, for each factor in $(p-1)!$, its inverse in $\mathbb{Z}_p$ is also a factor. In two cases, namely for the factors 1 and $p-1$, the inverse is the same factor (see Exercise 27), while in the other cases the inverse is a different factor. For $p \geq 3$ we see that

$$(p-1)! \equiv (p-1) \cdot \underbrace{(1)(1)\cdots(1)}_{\frac{p-3}{2}(1)'s} \cdot (1) \pmod{p}$$

so $(p-1)! \equiv p - 1 \equiv -1 \pmod{p}$. When $p = 2$, we have $p - 1 \equiv 1 \equiv -1 \pmod{2}$.

29. We show that $n^{37} - n$ is divisible by each of the primes 37, 19, 13, 7, 3, and 2 for every positive integer n. By Corollary 20.2, $a^p \equiv a \pmod{p}$ so of course $n^{37} \equiv n \pmod{37}$ for all n, so $n^{37} - n \equiv 0 \pmod{37}$ for all n, that is $n^{37} - n$ is divisible by 37 for all $n \in \mathbb{Z}+$.

Working modulo 19, we have $n^{37} - n \equiv n[(n^{18})^2 - 1]$. If n is divisible by 19, then so is $n^{37} - n$. If n is not divisible by 19, then by Fermat's theorem, $(n^{18})^2 - 1 \equiv 1^2 - 1 \equiv 0 \pmod{19}$, so again $n^{37} - n$ is divisible by 19. Notice that the reason this argumnet works is that 36 = 37 - 1 is a multiple of 18 = 19 - 1.

Divisiblity by 13, 7, 3, and 2 are handled in the same way, and the computatations are successful because

36 is a multiple of 13 - 1 = 12,
36 is a multiple of 7 - 1 = 6,
36 is a multiple of 3 - 1 = 2,
36 is a multiple of 2 - 1 = 1.

30. Looking at the argument in Exercise 29, we try to find still another prime p less than 37 such that 36 is divisible by $p - 1$. We see that $p = 5$ fills the bill, so $n^{37} - n$ is actually divisible by $5(383838) = 1919190$ for all integers n.

21. The Field of Quotients of an Integral Domain

1. The field of quotients of D is $\{q_1 + q_2 i \mid q_1, q_2 \in \mathbb{Q}\}$.

2. Because

$$\frac{1}{a+b\sqrt{2}} = \frac{1}{a+b\sqrt{2}} \cdot \frac{a-b\sqrt{2}}{a-b\sqrt{2}} = \frac{a}{a^2-2b^2} + \frac{-b}{a^2-2b^2}\sqrt{2},$$

we see that $\{q_1 + q_2\sqrt{2} \mid q_1, q_2 \in \mathbb{Q}\}$ is a field, and must be the field of quotients.

3. The definition is incorrect. We should think of the embedding as having taken place, so that $D \subseteq F$, and every element of F must be a quotient of elements of D.

A **field of quotients** of an integral domain D is a field F containing D as a subdomain and with the property that every $x \in F$ is equal to some quotient a/b for $a, b \in D$.

4. T F T F T T F T T T

5. Let $D = \{q \in \mathbb{Q} \mid q = m/2^n \text{ for } m, n \in \mathbb{Z}\}$, that is, the set of all rational numbers that can be written as a quotient of integers with denominator a power of 2. It is easy to see that D is an integral domain. Let $D' = \mathbb{Z}$. Then $\mathbb{Q}$ is a field of quotients of both D and D'.

6. We have

$$\begin{aligned}[(a,b)] + ([(c,d)] + [(e,f)]) &= [(a,b)] + [(cf+de, df)] \\ &= [(adf + bcf + bde, bdf)] \\ &= [(ad+bc, bd)] + [(e,f)] \\ &= ([(a,b)] + [(c,d)]) + [(e,f)].\end{aligned}$$

Thus addition is associative.

7. We have $[(0,1)] + [(a,b)] = [(0b+1a, 1b] = [(a,b)]$. by Part 1 of Step 3, we also have $[(a,b)] + [(0,1)] = [(a,b)]$.

8. We have $[(-a,b)] + [(a,b)] = [(-ab+ba, b^2)] = [(0,b^2)]$. But $[(0,b^2)] \sim [(0,1)]$ because $(0)(1) = (b^2)(0) = 0$. Thus $[(-a,b)] + [(a,b)] = [(0,1)]$. By Part 1 of Step 3, $[(a,b)] + [(-a,b)] = [(0,1)]$ also.

9. Now $[(a,b)]([(c,d)][(e,f)]) = [(a,b)][(ce,df)] = [(ace,bdf)] = [(ac,bd)][(e,f)] = ([(a,b)][(c,d)])[(e,f)]$. Thus multiplication is associative.

10. We have $[(a,b)][(c,d)] = [(ac,bd)] = [(ca,db)] = [(c,d)][(a,b)]$ so multiplication is commutative.

11. For the left distributive law, we have $[(a,b)]([(c,d)] + [(e,f)]) = [(a,b)][(cf+de,df)] = [(acf+ade,bdf)]$. Also, $[(a,b)][(c,d)] + [(a,b)][(e,f)] = [(ac,bd)] + [(ae,bf)] = [(acbf+bdae,bdbf)] \sim [(acf+ade,bdf)]$ because $(acbf+bdae)bdf = acbfbdf + bdaebdf = bdbf(acf+ade)$, for multiplication in D is commutative. The right distributive law then follws from Part 6.

12. **a.** Because T is nonempty, there exists $a \in T$. Then $[(a,a)]$ is unity in $Q(R,T)$, because $[(a,a)][(b,c)] = [(ab,ac)] \sim [(b,c)]$ since $abc = acb$ in the commutative ring R.

b. A nonzero element $a \in T$ is identified with $[(aa,a)]$ in $Q(R,T)$. Because T has no divisors of zero, $[(a,aa)] \in Q(R,T)$, and we see that $[(aa,a)][(a,aa)] = [(aaa,aaa)] \sim [(a,a)]$ because $aaaa = aaaa$. We saw in part **a** that $[(a,a)]$ is unity in $Q(R,T)$. Commutativity of $Q(R,T)$ shows that $[(a,aa)][(aa,a)]$ is unity also, so $a \in T$ has an inverse in $Q(R,T)$ if $a \neq 0$.

13. We need only takt $T = \{a^n \mid n \in \mathbb{Z}^+\}$ in Exercise 12. This construction is entirely different from the one in Exercise 30 of Section 19.

14. There are four elements, for 1 and 3 are already units in $\mathbb{Z}_4$.

15. It is isomorphic to the ring D of all rational numbers that can be expressed as a quotient of integers with denominator a power of 2, as described in the answer to Exercise 5.

16. It is isomorphic to the ring of all rational numbers that can be expressed as a quotient of integers with denominator a power of 6. The 3 in the $3\mathbb{Z}$ does not restrict the numerator, because 1 can be recovered as [(6, 6)], 2 as [(12, 6)], etc.

17. It runs into trouble when we try to prove the transitive property in the proof of Lemma 21.2, for multiplicative cancellation may not hold. For $R = \mathbb{Z}_6$ and $T = \{1,2,4\}$ we have $(1,2) \sim (2,4)$ because $(1)(4) = (2)(2) = 4$ and $(2,4) \sim (2,1)$ because $(2)(1) = (4)(2)$ in $\mathbb{Z}_6$, but $(1,2) \nsim (2,1)$ because $(1)(1) \neq (2)(2)$ in $\mathbb{Z}_6$.